

Integration

Origins and approaches

Key topics

Graphical interpretation & Trapezium approach

Notation

Definite Vs Indefinite

Addition and subtraction

Standard results

Definition

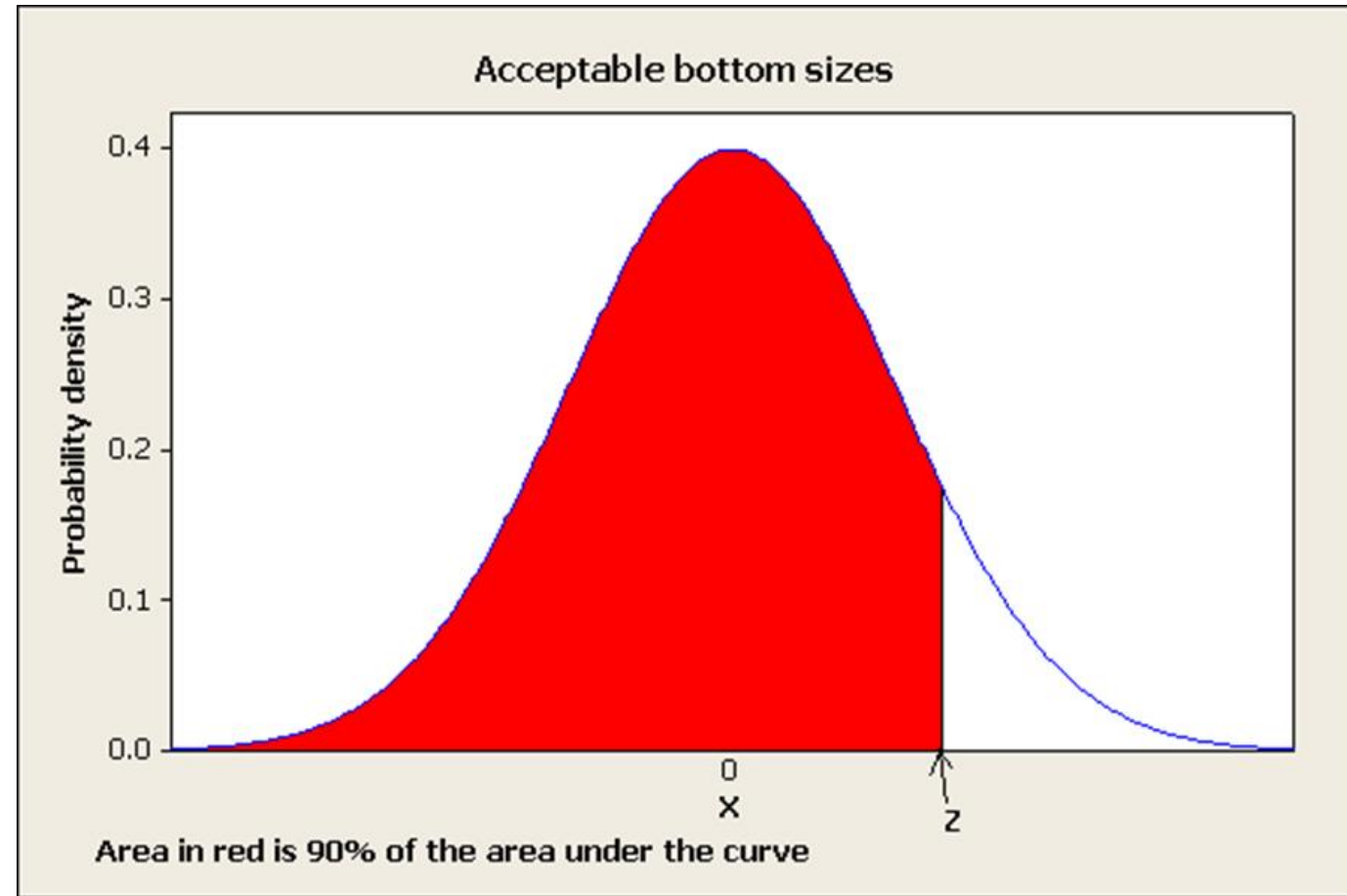
Integration is a technique used to find the area under a curve

Area under a curve can be:

A probability (e.g. Normal Distribution),

Drug absorption
(Pharmokinetics)

Displacement is the area under a velocity curve
(Physics)



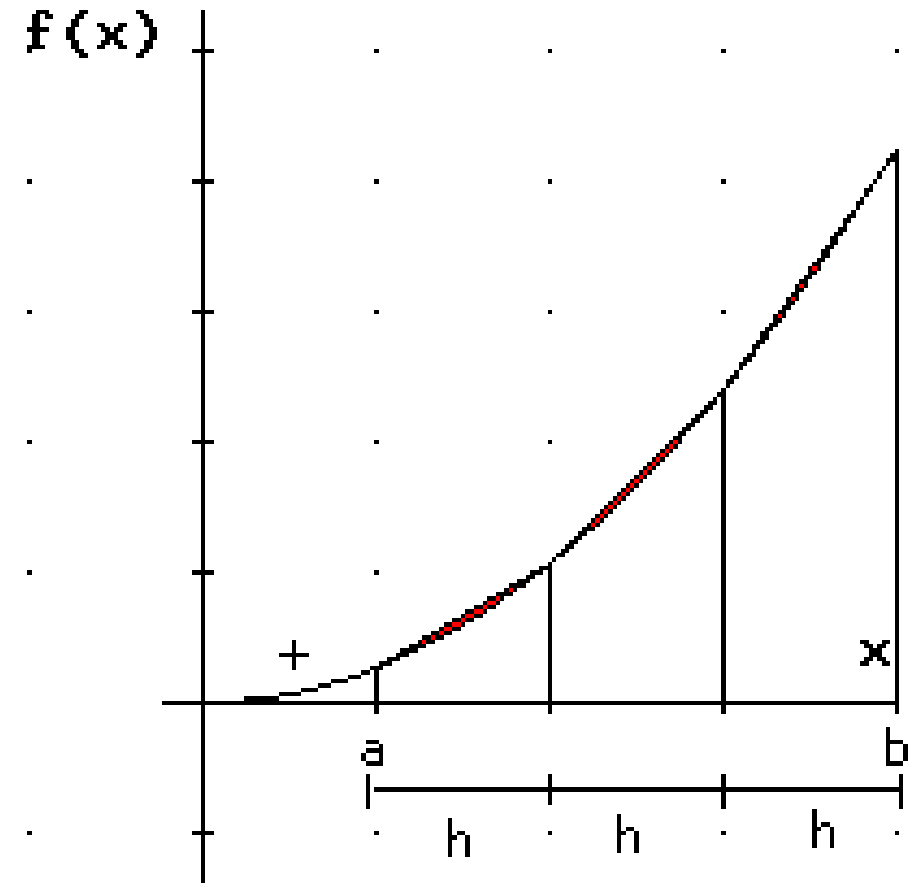
Numerical approach

Many numerical techniques used (left hand rule, right hand rule, Simpson's rule etc.)

We'll focus on the Trapezium rule:

Trapezium is a four sided shape with straight edges

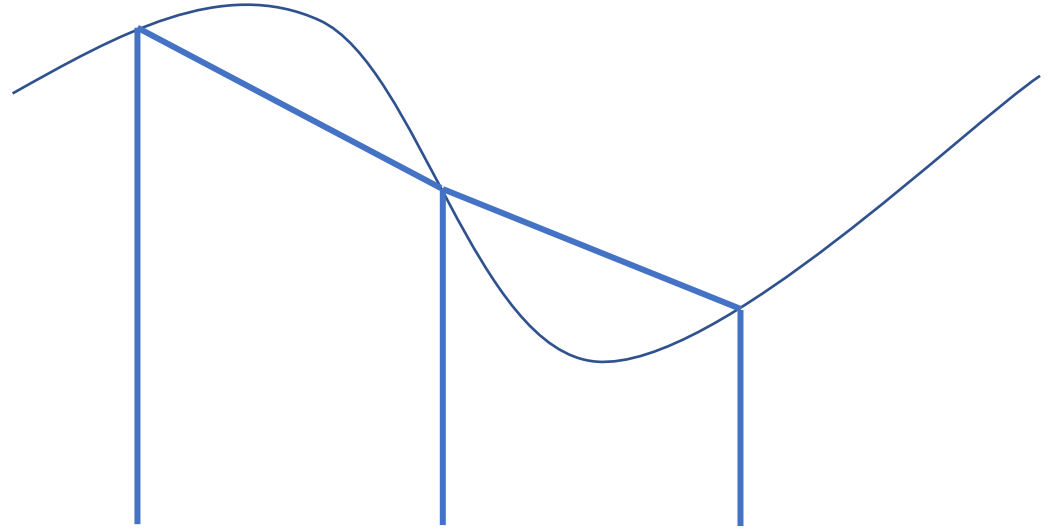
We try to estimate the area under a curve by fitting these shapes and counting them



Imperfections

We are approximating a curve using straight lines

Inevitably we will either over or under estimate

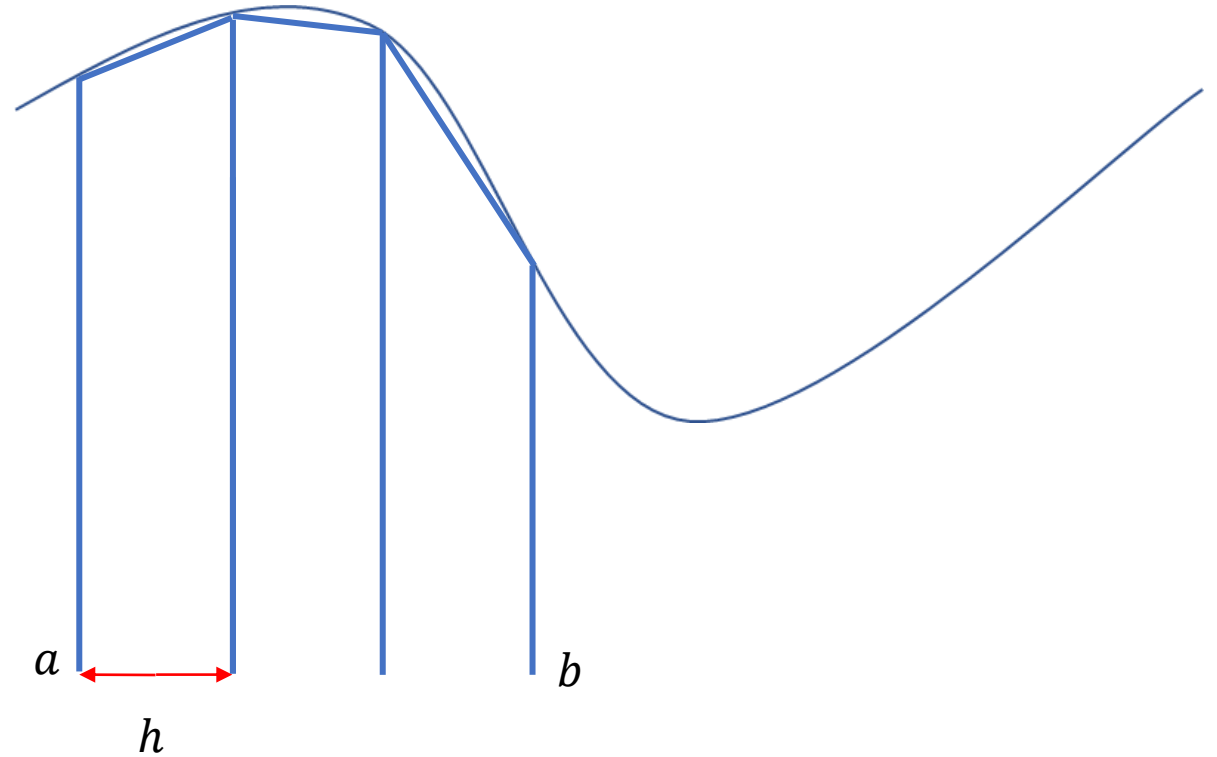


Improving accuracy

Decreasing the step-size, h
(the width of the trapezoids)
helps reduce our errors

$$h = \frac{b - a}{n}$$

Where n is the number of
steps and a, b are our end
points



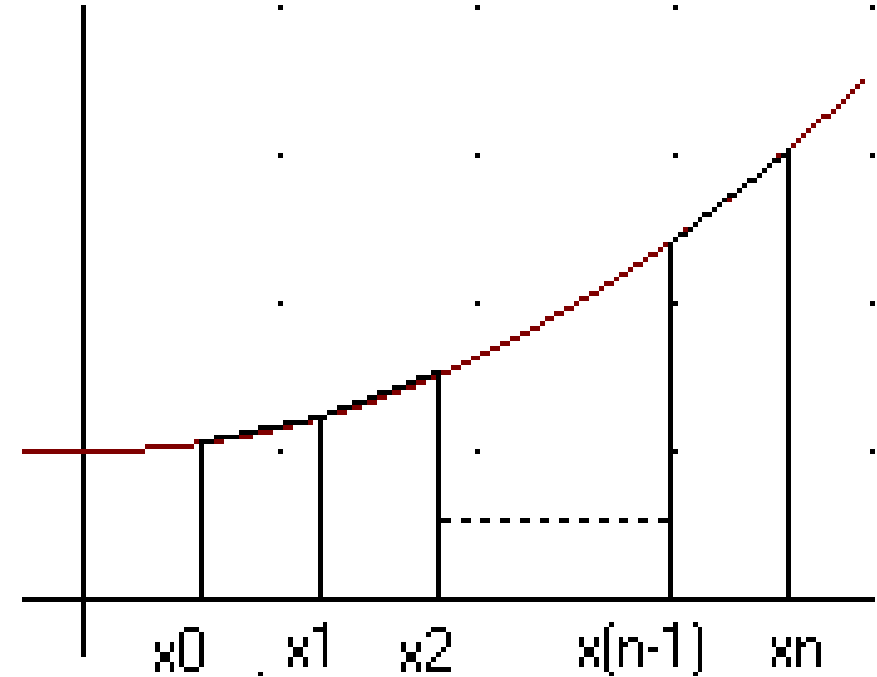
Defining the possibilities

Our worst case scenario is using a single strip (treating the curve as a straight line)

$$\frac{h}{2}(f(a) + f(b))$$

Adding in multiple strips gives (with $a = x_0, b = x_n$):

$$\frac{h}{2}(f(x_0) + f(x_1) + f(x_2) + \cdots + f(x_{n-1}) + f(x_n))$$



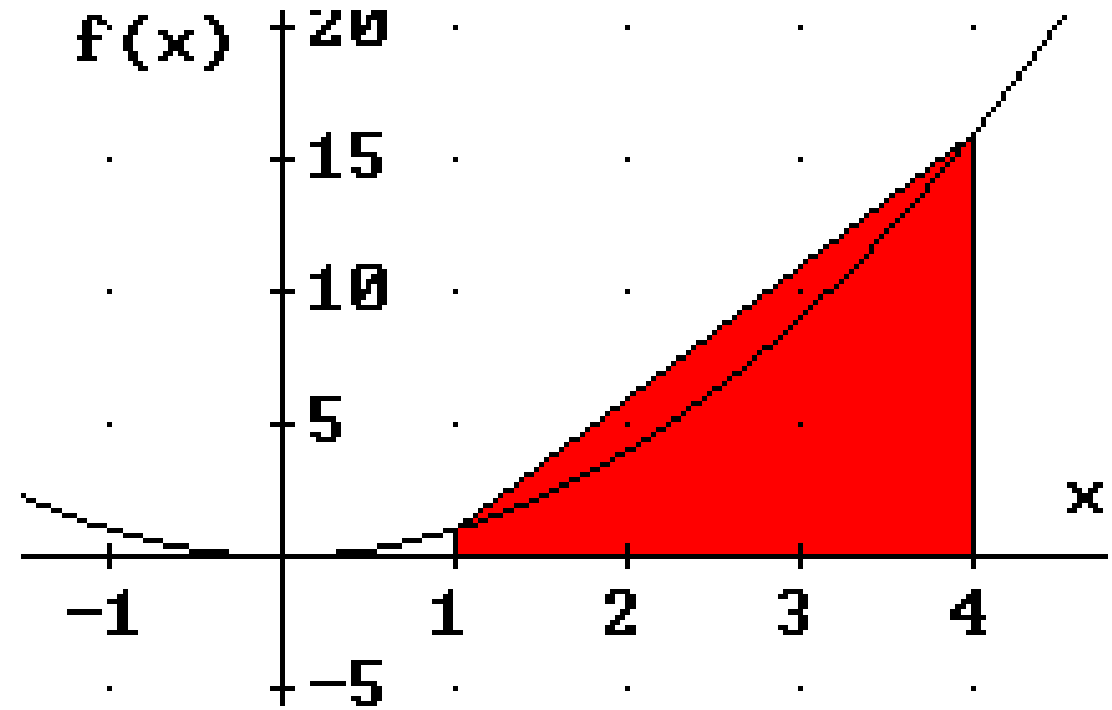
Single strip

Estimate the area under $f(x) = x^2$
between $x = 1$ and $x = 4$,

We have one strip: $n = 1$, so our
step size is: $h = 3$

$$Area = \frac{h}{2}(f(1) + f(4))$$

$$= \frac{3}{2}(1 + 16) = 25.5 \text{ units}^2$$

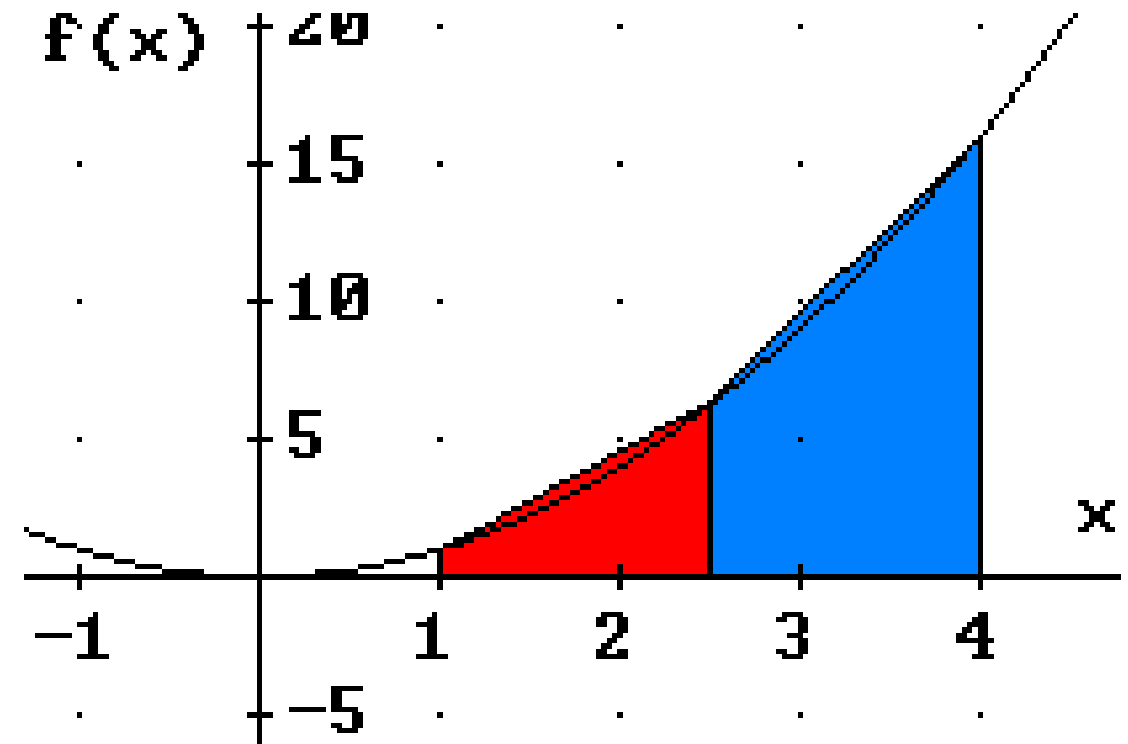


Double strip

Estimate the area under $f(x) = x^2$
between $x = 1$ and $x = 4$,

We have two strips: $n = 2$, so our step
size is: $h = 1.5$

$$\begin{aligned} \text{Area} &= \frac{h}{2}(f(1) + 2(f(2.5)) + f(4)) \\ &= \frac{1.5}{2}(1 + 12.5 + 16) = 22.125 \text{ units}^2 \end{aligned}$$



Accuracy

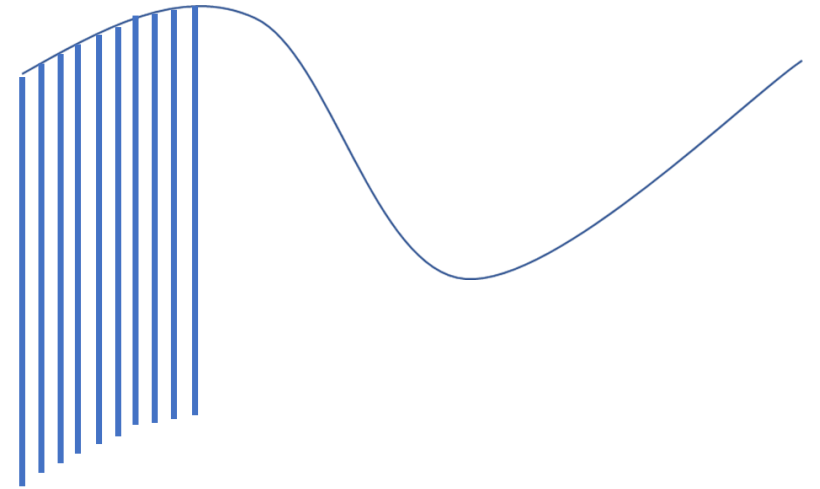
The true answer in this case is 21

1 strip: Error $\varepsilon = 25.5 - 21 = 4.5$

2 strips: Error $\varepsilon = 22.125 - 21 = 1.125$

As $n \rightarrow \infty, \varepsilon \rightarrow 0$

This is because the more strips we have, the more closely we approximate the **gradient** of the curve at every point



Infinite strips

We are going to want as many strips as possible, so that we can add them all up:

$$\sum_0^n width \times height$$

We used this summation sign, sigma, before when adding up all of the deviation terms for the standard deviation.

In fact, it is used for adding discrete ‘lumps’, but we are talking about adding up infinitesimal amounts, so instead we use the integral sign:

Discrete: $\sum_{start}^{end} trapeziums$

Continuous: $\int_{start}^{end} f(x)$

Integration formula

The integral of a function defines the area under the curve exactly.

As with differentiation, there is a formula for finding the integral of a function:

$$y = 5x^3 + 3$$

$$\frac{dy}{dx} = 15x^2$$

$$\int y = \frac{5}{4}x^4$$

The general solution

For differentiation, we increased the coefficient and reduced the power.

we had that : ax^n becomes anx^{n-1}

For integration, we do the opposite.

we have that: ax^n becomes $\frac{a}{n+1}x^{n+1}$

Notation

When we differentiated, we used this symbol to show which variable we were differentiating with respect to (usually x):

$$\frac{dy}{dx}$$

We do something similar (but opposite!!!) when integrating:

$$\int f(x)dx$$

Moving between integrals and differentials

It looks like we are reversing the process of differentiation, and that's nearly correct.

$$y = 5x^3 + 3$$

$$\frac{dy}{dx} = 15x^2$$

$$\int 15x^2 dx = 5x^3 + \text{?????}$$

What we are missing is the constant!

degeneracy

A degenerate problem has multiple solutions:

$$y_1 = 3x^2, \quad y_2 = 3x^2 + 1, \quad y_3 = 3x^2 - 4$$

$$\frac{dy_1}{dx} = 6x, \quad \frac{dy_2}{dx} = 6x, \quad \frac{dy_3}{dx} = 6x$$

$$\int 6x \, dx = 3x^2 \quad \int 6x \, dx = 3x^2 \quad \int 6x \, dx = 3x^2$$

Because we might have any number of original functions,
we add an arbitrary constant, c

$$c_1 = 0 \quad c_2 = 1 \quad c_3 = -4$$

General Vs Specific

The general solution to: $\int 3x^2 dx$ is: $\int 3x^2 = x^3 + c$

For us to get a specific answer we need more information. We need to know some point that the original function passed through.

What is y when $\frac{dy}{dx} = 12x^3$, given that $y = 5$ when $x = 1$?

$$\int 12x^3 dx = \frac{12}{4}x^4 + c = 3x^4 + c$$

$$5 = 3(1)^4 + c, \text{ so } c = 2$$

Example

$$\int ax^n dx = \left(\frac{a}{n+1} \right) x^{n+1} + c$$

What is the integral of $10x^4$?

$$\int 10x^4 dx = \left(\frac{10}{4+1} \right) x^{4+1} + c = 2x^5 + c$$

Now your turn, try integrating: $\frac{1}{x}$

A special case

$$\int \frac{1}{x} dx$$

$$= \int x^{-1} dx$$

$$= \frac{1}{0} x^0 + c$$

We have a problem, we can't divide by zero.

This gives us an exception to the general formula.

In fact, we recall:

$$\frac{dy}{dx} \ln x = \frac{1}{x}$$

So:

$$\int x^{-1} dx = \ln x + c$$

List of Standard integrals

Function

Integral

x

$\frac{1}{2}x^2$

x^n

$\frac{1}{n+1}x^{n+1}$

$\frac{1}{x}$

$\ln(x)$

e^x

e^x

e^{kx}

$\frac{1}{k}e^{kx}$

$\sin(x)$

$-\cos(x)$

$\cos(x)$

$\sin(x)$

$\sin(kx)$

$-\frac{\cos kx}{k}$

$\cos(cx)$

$\frac{\sin cx}{c}$

$\tan x$

$\frac{c}{\ln(\sec x)}$

Integration addition

$$\int (f(x) + g(x))dx = \int f(x)dx + \int g(x)dx$$

$$\int (x^2 + x^3)dx = \int x^2 dx + \int x^3 dx$$

$$\int (x^2 + x^3)dx = \frac{x^3}{3} + c_1 + \frac{x^4}{4} + c_2$$

$$\int (x^2 + x^3)dx = \frac{x^3}{3} + \frac{x^4}{4} + c$$

Integration subtraction

$$\int (f(x) - g(x))dx = \int f(x)dx - \int g(x)dx$$

$$\int (x^2 - x^3)dx = \int x^2dx - \int x^3dx$$

$$\int (x^2 - x^3)dx = \frac{x^3}{3} + c_1 - \frac{x^4}{4} - c_2$$

$$\int (x^2 - x^3)dx = \frac{x^3}{3} - \frac{x^4}{4} + c$$

Constant multiples

$$\int k \times f(x) \, dx = k \int f(x) \, dx$$

$$\int 3x^4 \, dx = 3 \int x^4 \, dx$$

$$\int 3x^4 \, dx = 3 \frac{x^5}{5} + 3c$$

$$\int 3x^4 \, dx = 3 \frac{x^5}{5} + c$$

Example

Find the indefinite integral:

$$\int 4x - 2x^2 + 8 \, dx$$

$$2 \int 2x - x^2 + 4 \, dx$$

$$2 \left(\frac{2x^2}{2} - \frac{x^3}{3} + 4x + c \right)$$

$$2 \left(x^2 - \frac{x^3}{3} + 4x + c \right) \text{ or } 2x^2 - \frac{2x^3}{3} + 8x + c$$

Definite values

We found a way to describe the gradient of a curve at any point, using differentiation:

$$y = 3x^2, \quad \frac{dy}{dx} = 6x$$

But we needed to include the values for the point we were interested in, e.g. What is the slope at $x = 4$?

$$\frac{dy}{dx} = 6x = 6(4) = 24$$

We can do the same with Integration, to find the area under the curve over a certain range. Because a slope is a one-dimensional construct and an area is a two dimensional one, we need to enter two values

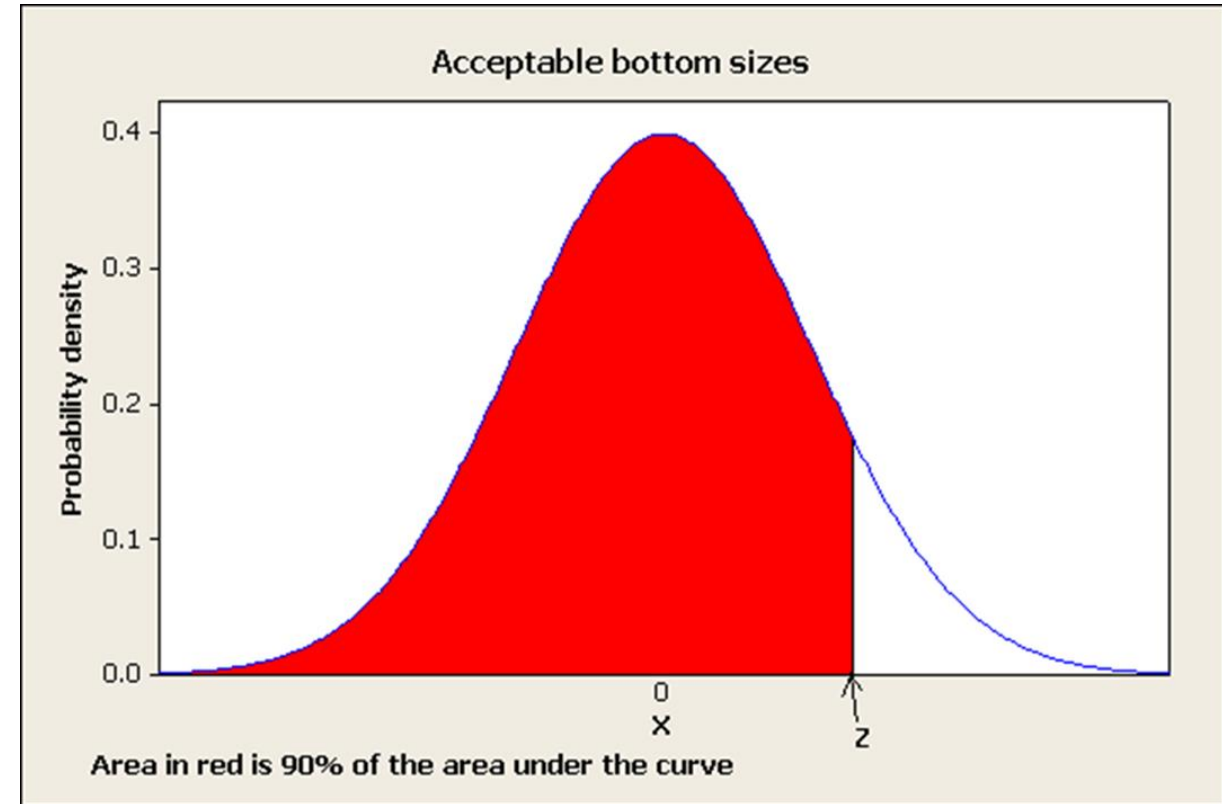
Definite integrals

So far we have used: $\int f(x)dx$,
This is the indefinite integral.

Its value is not 'defined' because we have not yet decided what limits to use for finding our area (remember that's what we are doing?)

In the case to the right, we have limits of –infinity and z , so we write:

$$\int_{-\infty}^z f(x)dx$$



Limits

Indefinite: $\int ax^n dx$

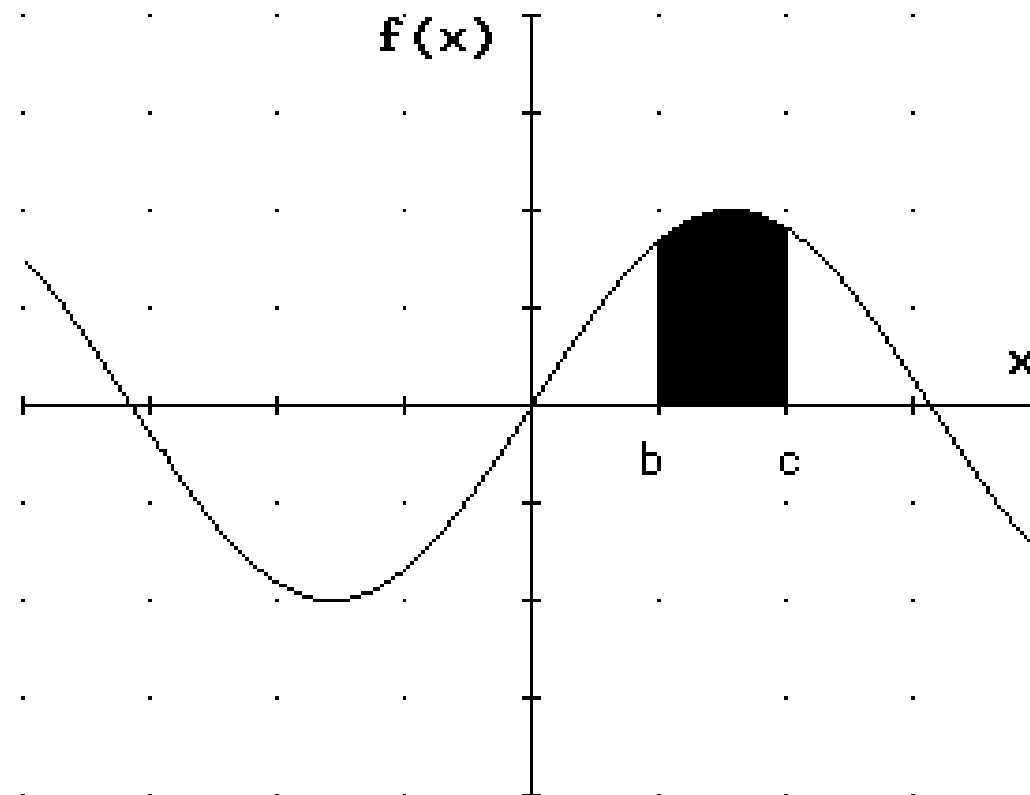
Definite: $\int_b^c ax^n dx$

We can move the constant outside the integral:

Indefinite: $\int ax^n = a \int x^n dx$

Definite: $\int_b^c ax^n = a \int_b^c x^n dx$

$$= a \left(\frac{1}{n+1} c^{n+1} - \frac{1}{n+1} b^{n+1} \right) = \frac{a}{n+1} (b^{n+1} - c^{n+1})$$



Limits example

We can take out the constant

$$\int_1^3 2x^5 dx$$

Then integrate the expression

$$2 \int_1^3 x^5 dx$$

Then find the difference
between the top limit and the
bottom limit

$$2 \left[\frac{x^6}{6} + c \right]_1^3$$

$$\frac{(\mathbf{3})^6}{3} + c - \left(\frac{(\mathbf{1})^6}{3} + c \right)$$

The c terms cancel out

$$243 + c - \frac{1}{3} - c = 242.\dot{6}$$

Limits example, your turn

Evaluate the definite integral:

$$\int_{-1}^1 3x^2 + 7 \, dx$$

$$[x^3 + 7x]_{-1}^1$$

$$(1)^3 + 7(1) - (-1)^3 - 7(-1)$$

$$1 + 7 + 1 + 7 = 16$$

Negative areas

This is the function:

$$y = x^3$$

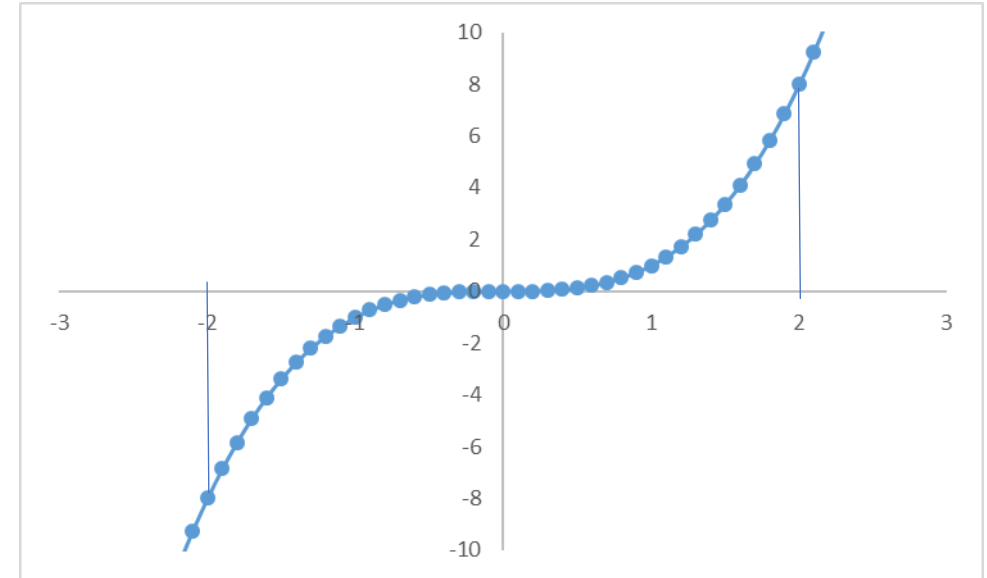
Consider integrating from -2 to 2

$$\int_{-2}^2 x^3 dx = \left[\frac{x^4}{4} \right]_{-2}^2$$

$$= \frac{16}{4} - \left(\frac{16}{4} \right) = 0$$

The x axis is what we are integrating up to, it is like a limit

Negative or zero answers are possible
Take care interpreting them



Further help

More examples in Foundation Maths by Tony Croft: Chapter 36, p462